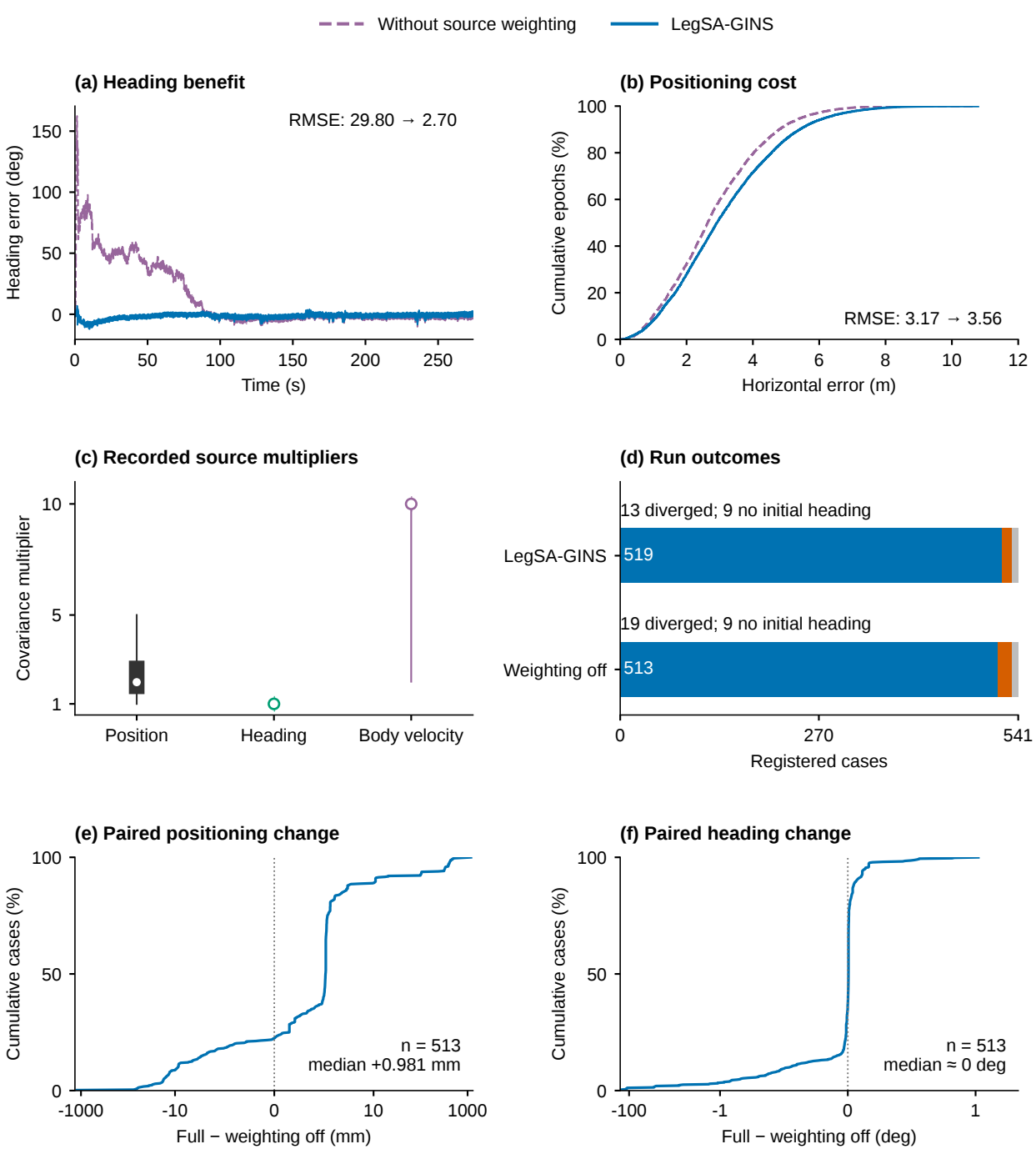

Without source weighting
LegSA-GINS
(a) Heading benefit
RMSE: 29.80 → 2.70
Heading error (deg)
Time (s)
(b) Positioning cost
Cumulative epochs (%)
RMSE: 3.17 → 3.56
Horizontal error (m)
(c) Recorded source multipliers
Covariance multiplier
Position
Heading
Body velocity
(d) Run outcomes
13 diverged; 9 no initial heading
LegSA-GINS
519
19 diverged; 9 no initial heading
Weighting off
513
Registered cases
(e) Paired positioning change
Cumulative cases (%)
n = 513
median +0.981 mm
Full − weighting off (mm)
(f) Paired heading change
Cumulative cases (%)
n = 513
median ≈ 0 deg
Full − weighting off (deg)